BANKING CUSTOMER & TRANSACTION MANAGEMENT SYSTEM

Professional SQL Project Documentation

Technology	Project Type	Records
MySQL 8.0	Data Analytics / SQL	780

Purpose: Build an end-to-end relational banking database and use SQL to validate data, analyze customers/accounts/transactions/loans, calculate business KPIs, identify customer and branch patterns, and generate actionable business insights.

1. Executive Summary

This project demonstrates an end-to-end SQL analytics workflow for a banking environment. A relational database was designed for customers, accounts, branches, transactions, loans, cards, and beneficiaries. The project then progressed from data creation and validation to business analysis, advanced SQL, KPIs, customer segmentation, and business insights.

The solution uses MySQL 8.0 features including joins, aggregations, subqueries, CASE expressions, CTEs, window functions, views, ranking, running totals, and period-over-period analysis. The final analysis contains 50 business-insight queries covering engagement, product adoption, transaction behavior, loan exposure, digital usage, branch performance, and customer segments.

2. Project Objective

- Design a realistic banking relational database using normalized tables and foreign-key relationships.
- Create and validate a synthetic banking dataset directly in MySQL.
- Perform customer, account, transaction, branch, and loan analysis.
- Demonstrate advanced SQL concepts used in Data Analyst interviews.
- Translate SQL results into business KPIs, findings, and recommendations.
- Create a portfolio-ready project that can be explained clearly in interviews.

3. Business Problem

A bank needs a structured way to understand customer accounts, balances, transaction behavior, loans, cards, beneficiaries, and branch-level performance. Raw operational data alone does not answer business questions such as which customers are highly valuable, where transaction activity is concentrated, which accounts have low engagement, how digital channels are used, and which customers have significant loan exposure.

This project addresses those questions using SQL-based analysis over a connected banking dataset.

4. Technology & Environment

- Database: MySQL 8.0.43
- SQL capabilities: DDL, DML, joins, aggregate functions, subqueries, CASE, CTEs, window functions, views, EXISTS/NOT EXISTS, UNION ALL, COALESCE/IFNULL.
- Data volume: 780 records across 7 core tables.
- Analysis focus: Data Analyst / SQL interview-oriented business analysis.

5. Database Architecture

The database is named **banking_management_system**. The design uses customers and branches as core entities, with accounts acting as a central operational entity. Transactions, cards, and beneficiaries connect to accounts, while loans connect directly to customers.

Core relationships

Parent	Child	Cardinality
customers.customer_id	accounts.customer_id	1 : Many
branches.branch_id	accounts.branch_id	1 : Many

Parent	Child	Cardinality
accounts.account_id	transactions.account_id	1 : Many
customers.customer_id	loans.customer_id	1 : Many
accounts.account_id	cards.account_id	1 : Many
accounts.account_id	beneficiaries.account_id	1 : Many

6. Tables & Dataset

Table	Purpose	Rows
branches	Branch master data, city/state, IFSC	20
customers	Customer demographic and registration data	50
accounts	Customer accounts, branch, type, balance, status	70
transactions	Financial transaction activity and channels	500
loans	Customer loan exposure and status	30
beneficiaries	Account beneficiary details	50
cards	Account card details and status	60
Total		**780**

Dataset creation was completed directly in MySQL, including generation of 500 transaction records. This keeps the project reproducible without requiring external CSV imports.

7. Data Validation & Quality Checks

- Verified row counts for all seven tables.
- Checked duplicate primary-key values.
- Checked NULL values in relevant fields.
- Checked foreign-key orphan records.
- Validated transaction amounts and loan amounts.
- Checked for negative account balances.
- Confirmed expected relationships between parent and child tables.

Validation outcome: the expected checks passed for the created synthetic dataset.

8. SQL Analysis Performed

The project moved from simple descriptive analysis to advanced business-oriented analysis.

8.1 Basic SQL Analysis

- Total customers, accounts, and bank balance.
- Average account balance and balance by account type.
- Customers by gender, state, and city.
- Transaction count and amount by transaction type and mode.
- Loan count and amount by loan type and status.

- Account count and balance by branch.

8.2 Customer Analysis

- Customers with multiple accounts.
- Top customers by balance.
- Customers with and without accounts or loans.
- Customer account and loan summaries.
- Customer transaction summaries.
- High-balance and high-activity customer identification.

8.3 Account Analysis

- Account status and account-type analysis.
- Top accounts by balance.
- Average balance by account type.
- Accounts opened by year/month.
- Low-balance and inactive account identification.

8.4 Transaction Analysis

- Transaction analysis by type, mode, date, and account.
- Daily/monthly transaction trends.
- Top accounts and customers by transaction activity.
- Accounts with no transactions.
- Digital channel adoption and multi-channel behavior.

8.5 Branch Analysis

- Branch account and balance performance.
- Branch transaction performance.
- Branch customer-to-account ratio.
- Branch account-type distribution.
- Branch loan-related analysis.

8.6 Loan Analysis

- Loan count and exposure by type.
- Loan status analysis.
- Highest loan-exposure customers.
- Loan amount by interest rate.
- Loan distribution by customer state.

9. Advanced SQL Concepts Demonstrated

- Subqueries and correlated subqueries for above-average and conditional analysis.
- EXISTS and NOT EXISTS for presence/absence checks.
- IN and UNION ALL for multi-condition analysis.
- CASE expressions for classification and segmentation.
- COALESCE and IFNULL for null-safe calculations.
- GROUP BY and HAVING for aggregated business conditions.
- CTEs for readable multi-stage analytical logic.
- Window functions for ranking, running totals, customer-wise calculations, and period comparisons.
- ROW_NUMBER, RANK, and DENSE_RANK for ranking use cases.
- LAG and LEAD for month-over-month and sequential analysis.
- SQL Views for reusable analytical datasets.

10. CTE & Window Function Analysis

CTEs were used to separate complex calculations into logical stages, especially for customer summaries and top-customer analysis. Window functions were used to calculate rankings, running totals, customer-level partitions, monthly changes, growth percentages, and top accounts within each branch.

Examples of analytical patterns include customer-wise transaction ranking, cumulative transaction totals, month-over-month comparison, and top-three accounts per branch.

11. SQL Views

View	Purpose
customer_account_summary	Reusable customer-level account summary
customer_transaction_summary	Reusable customer-level transaction summary
branch_performance	Reusable branch performance metrics
loan_performance	Reusable loan performance metrics

Views make frequently reused business logic easier to query and demonstrate practical SQL engineering beyond one-off SELECT statements.

12. Business KPI Framework

- Overall banking KPIs: customers, accounts, total balance, transaction activity, and loan exposure.
- Active customer rate.
- Average balance per customer.
- Transaction activity rate.
- Average transaction value.
- Loan-to-balance ratio.
- Active loan rate.
- Loan penetration rate.
- Digital transaction rate.

- Branch performance KPIs.
- Transaction volume by channel.
- Account-type contribution.
- Customer segmentation by balance.

These KPIs convert raw database records into metrics that can be discussed from a business perspective during an interview.

13. Business Insights Framework

A dedicated set of 50 business-insight queries was developed. The analysis covers the following themes:

- High-value customers with low transaction activity.
- Customers with multiple accounts and low balances.
- High-value customers with high transaction activity.
- Branches with high account counts but comparatively low balance.
- Digital transaction adoption and high digital usage.
- Most active branches.
- High-balance accounts/customers with no transactions.
- High transaction value with low transaction count.
- High transaction activity with low average branch balance.
- Multiple-account customers with high activity.
- High balance combined with active loans.
- Highly active customers with no loans.
- High balance combined with low digital usage.
- Multi-product customer adoption across accounts, cards, loans, and beneficiaries.
- Customers with accounts but no cards/transactions/loans.
- Customers with cards but no loans, or loans but no cards.
- Multi-channel transaction behavior.
- High loan exposure combined with low balance.
- Customers with high balance but no active loans.
- Branch customer count versus branch balance.
- Low product penetration.
- Average balance and average loan value by branch.
- Card status and multiple-card analysis.
- Beneficiary adoption.
- High transaction activity combined with cards or loans.
- Inactive accounts and recently opened accounts with low activity.
- Final customer engagement scoring using multiple product and activity dimensions.

Technical note: where multiple one-to-many tables were combined, some exploratory queries used DISTINCT-based aggregation to reduce join multiplication. In production analytics, pre-aggregating each child

table in separate CTEs before joining is generally safer when duplicate numeric values can legitimately occur.

14. Key Findings & Interpretation

The project is designed to surface patterns rather than make unsupported assumptions about customer behavior. Findings should be interpreted against the synthetic dataset and business thresholds used in each query.

- Customer value can be evaluated using a combination of balance, transaction activity, and product adoption rather than a single metric.
- Low transaction activity among high-balance customers can identify accounts that warrant engagement analysis.
- Digital transaction usage can be evaluated by comparing digital modes such as UPI, IMPS, NEFT, and RTGS with other channels.
- Branch performance can be viewed through multiple dimensions including customers, accounts, balances, transactions, and loans.
- Product penetration across accounts, cards, loans, and beneficiaries provides a broader view of customer engagement.
- Loan exposure should be reviewed alongside customer balance and transaction behavior to identify different customer profiles.
- Inactive or recently opened accounts with low activity can be separated from established high-activity accounts.

15. Business Recommendations

- Use multi-dimensional customer segmentation for relationship-management campaigns.
- Investigate high-balance but low-activity accounts for engagement opportunities.
- Encourage relevant digital transaction channels where adoption is low.
- Review branches using a balanced set of customer, balance, and transaction KPIs rather than a single measure.
- Monitor customers with significant loan exposure relative to available account balances.
- Track inactive and recently opened accounts separately so normal onboarding behavior is not confused with long-term inactivity.
- Use product-penetration analysis to identify opportunities for relevant cross-product engagement.

16. Project Workflow

Phase	Work Completed
1. Database	Created banking_management_system database
2. Schema	Designed 7 relational tables and foreign keys
3. Data	Inserted 780 synthetic records
4. Validation	Checked duplicates, NULLs, orphan records, amounts, balances
5. Analysis	Performed customer/account/transaction/branch/loan analysis
6. Advanced SQL	Applied subqueries, CASE, EXISTS, CTEs, window functions
7. Reusability	Created analytical SQL views
8. KPI Layer	Calculated banking and customer metrics
9. Insights	Built 50 business-oriented insight queries
10. Documentation	Consolidated the project for portfolio/interview use

17. Technical Notes & Limitations

- The dataset is synthetic and intended for learning, portfolio, and interview demonstration purposes.
- The transaction records were generated for controlled SQL practice and do not represent real bank activity.
- Loans are linked to customers rather than branches; therefore branch loan attribution requires a clear business rule in a real banking system.
- For production reporting, child-table metrics should generally be pre-aggregated before multi-table joins to avoid row multiplication.
- Thresholds used in business-insight queries should be parameterized and agreed with business stakeholders in a real implementation.

18. Conclusion

The Banking Customer & Transaction Management System demonstrates a complete SQL analytics project from relational database design through business insight generation. It combines core SQL fundamentals with advanced analytical techniques and converts database records into meaningful banking KPIs and customer/branch insights.

The project is suitable as a Data Analyst portfolio project because it demonstrates practical SQL problem solving, relational thinking, data validation, analytical querying, business interpretation, and interview-ready advanced SQL concepts.